VoiceOps 2026 Capstone Report

Audit snapshot: 25 October 2026 · Dataset reference date: 20 October 2026

A four-step support agent: Gemini prompting, document-grounded RAG with source IDs, multi-turn conversations with verified order and ticket tools, and Gnani-only speech. PowerShell, local website and hosted website share the same Python agent. All customer records are synthetic.

Saved full-pack evaluation

Run	Scope	Required facts	Evidence / tools
Step 1 · prompt only	52 questions	8 / 49 (16.3%)	No RAG; tool screening 52 / 52
Step 2 · grounded RAG	52 questions	42 / 49 (85.7%)	Evidence 47 / 47; tools 52 / 52
Step 3 · single questions	52 questions	40 / 49 (81.6%)	Evidence 47 / 47; tools 52 / 52
Step 3 · ASR-text simulation	52 questions	35 / 49 (71.4%)	Evidence 44 / 47; tools 52 / 52
Step 3 · conversations, post-fix	96 turns	59 / 77 (76.6%)	Evidence 96 / 96; tools 96 / 96

Interpretation. Saved full-pack Step 1/2 scores use the same `question` field; they are historical results, not a fresh full run on `asr_text`. Automated checks can miss valid paraphrases and require human review.

Quota-limited live asr_text pilot

Run	Facts	Gold evidence	Other checks
Step 1 baseline	1 / 3	0 / 2	Prompt-only
Step 2 before fix	1 / 3	1 / 2	Initial sample
Step 2 after fix	3 / 3	2 / 2	Hard negatives 3 / 3; tools 3 / 3

Cases: PQ0002 spoken-SKU warranty, PQ0003 wallet refund timing, PQ0004 unsupported midnight delivery. The three-case pilot is a smoke sample, not a benchmark; Gemini requests used 4.1-second spacing.

Verified fixes

Observed issue	Targeted change	Verified result
Spoken SKU did not map	Normalize spoken digits against actual SKUs	PQ0002 retrieved ELE-0012 and WAR-2; scored facts and evidence passed
Unsupported delivery slot guess	Clarify lead-time does not establish slot availability	PQ0004 abstained instead of inventing hours
Gnani WAV length sentinel	Measure actual RIFF data length	Synthetic batch passed; reply WAV measured 13.44 seconds

Voice, verification & next steps

Live speech test and reproducibility snapshot

Voice and offline verification

Check	Verified result
Synthetic audio batch	One synthetic WAV completed Gnani STT → Gemini → Gnani TTS. Transcript, reply, 11 retrieved chunks and 13.44-second reply audio are saved under sample_outputs/live_voice_smoke_verified/.
Live website microphone	Gnani transcribed “Where is my order?”; Gemini gave the expected request for an eight-digit order ID. Sources appeared in the browser and a six-second Gnani reply was available. Personal recording not retained.
Offline tests	38 tests: 37 passed; one socket-dependent test skipped because local sockets are blocked.
Line coverage	81.1% (1,422 / 1,754 executable lines), 10 first-party Python modules; standard-library trace. Branch coverage not measured.
Dataset validator	Passed all nine CSVs, 21 knowledge documents, joins, order math, practice evidence and verification gates.

Interpretation and remaining work

The three-case live pilot supports the targeted fixes but cannot establish full-set performance. The saved 96-turn conversation run passed all tool-call and evidence checks; required-fact checks remain a screening signal. Full Step 1/2 `asr_text` evaluation and independent native-speaker review remain outstanding. No full API-consuming evaluation was run for this final check.

Reproduce locally

Purpose	Command
Install	`python -m pip install -r requirements.txt`
Validate data	`python validate_dataset.py`
Interactive agent	`python run.py --step 3`
Local website	`python web_app.py → http://127.0.0.1:8000`
Audio folder	`python batch_run.py --output sample_outputs`
Small Gemini sample	`python evaluate.py --dataset dataset --steps 1 2 --limit 3 --output reports/asr-pilot.jsonl`

Keep credentials private. Configure keys in the ignored local `.env`. Dataset copies, API keys, virtual environments and the personal mic recording are excluded from the submission ZIP.

GitHub: https://github.com/vanshdixit1/zentromart-voiceops · Live: https://zentromart-voiceops.vercel.app · Text agent: python run.py --step 3